PRACTICAL LAB REPORT

WEEK 1
PROJECT MODULE 1

VirtualBox + Kali Linux
Cybersecurity Lab Setup

Prepared by	Yousra
Host system	Windows
Virtualization platform	Oracle VirtualBox 7.2.22
Guest operating system	Kali Linux

OVERVIEW

This exercise established a local virtual environment for cybersecurity practice. The work moved from preparing the virtualization platform to configuring the Kali guest, checking its network interface, and making the host Downloads directory accessible from inside the VM.

TARGET CONFIGURATION

The practical specification required a NAT Network on 10.0.0.0/24, Kali Linux address 10.0.0.2/24, clipboard and drag-and-drop integration, a host Downloads shared folder, Internet connectivity, and a saved VM snapshot.

SOFTWARE

Windows host · Oracle VirtualBox 7.2.22 · Kali Linux guest · VirtualBox network and shared-folder controls

01 | PREPARING THE VIRTUALIZATION PLATFORM

The host system was prepared by installing VirtualBox 7.2.22. The completed installer and the VirtualBox Manager screen show that the platform was available for creating the Kali guest environment.



Figure — VirtualBox setup completed.

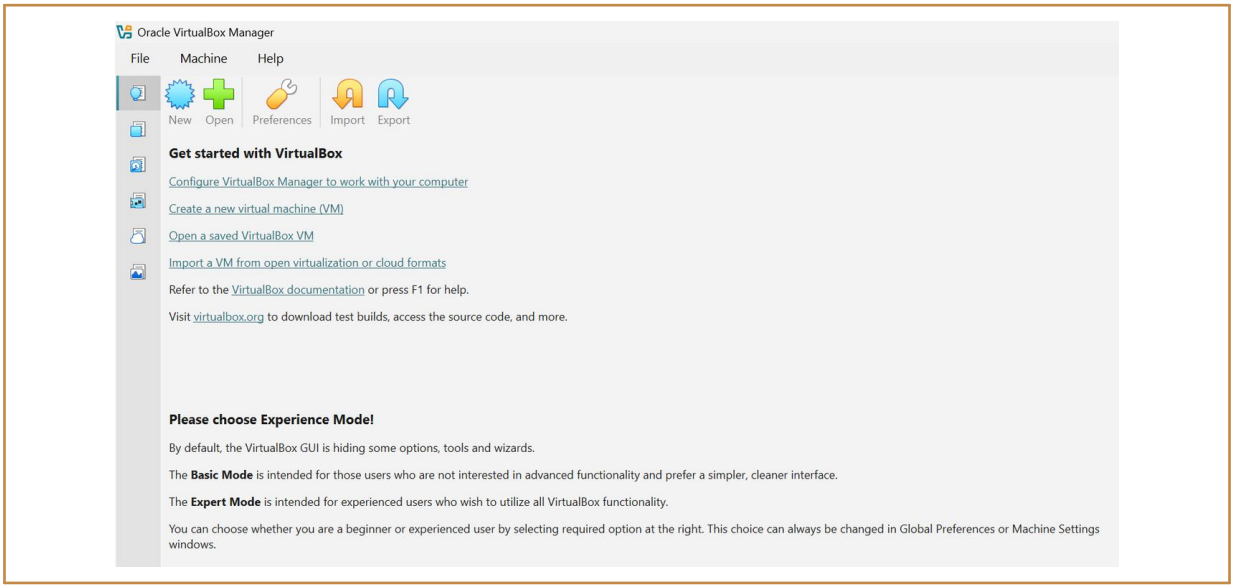


Figure — Manager window ready.

02 | DEFINING THE KALI GUEST SYSTEM

A virtual machine labelled KALI was configured in the manager. The setup wizard was used to define the guest machine and allocate its virtual hardware and disk before proceeding with the operating-system installation.

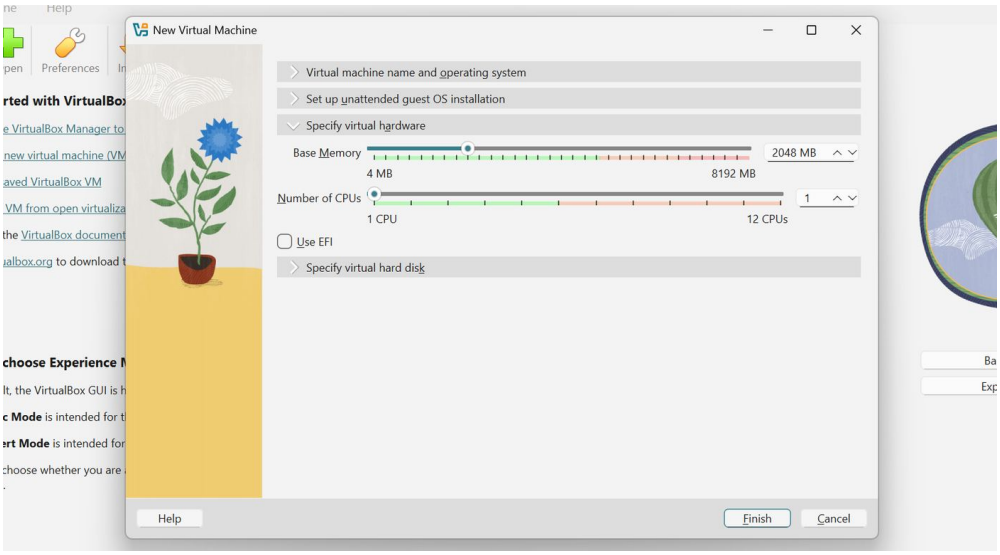


Figure — Virtual machine hardware details.

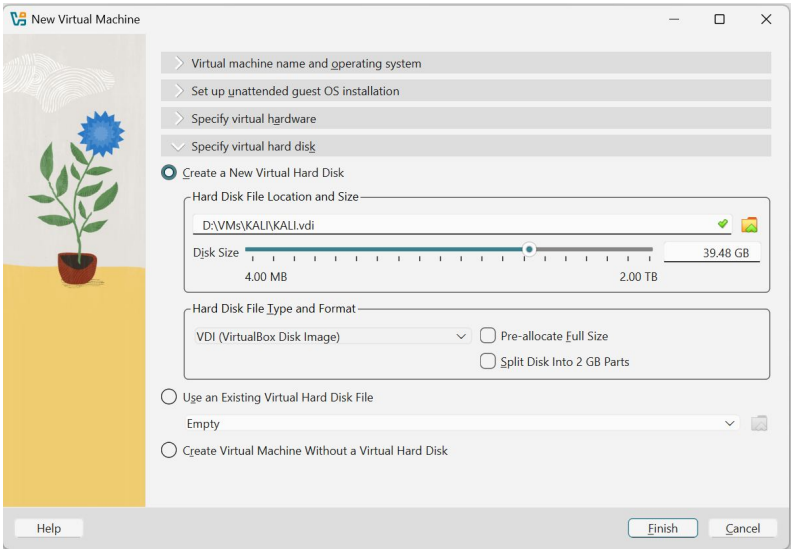


Figure — Virtual disk configuration.

03 | CONNECTING THE VM TO NATNETWORK

The virtual machine's network adapter was set to the NAT Network used for the exercise. The selected adapter configuration provides the network connection required for the 10.0.0.0/24 lab topology.

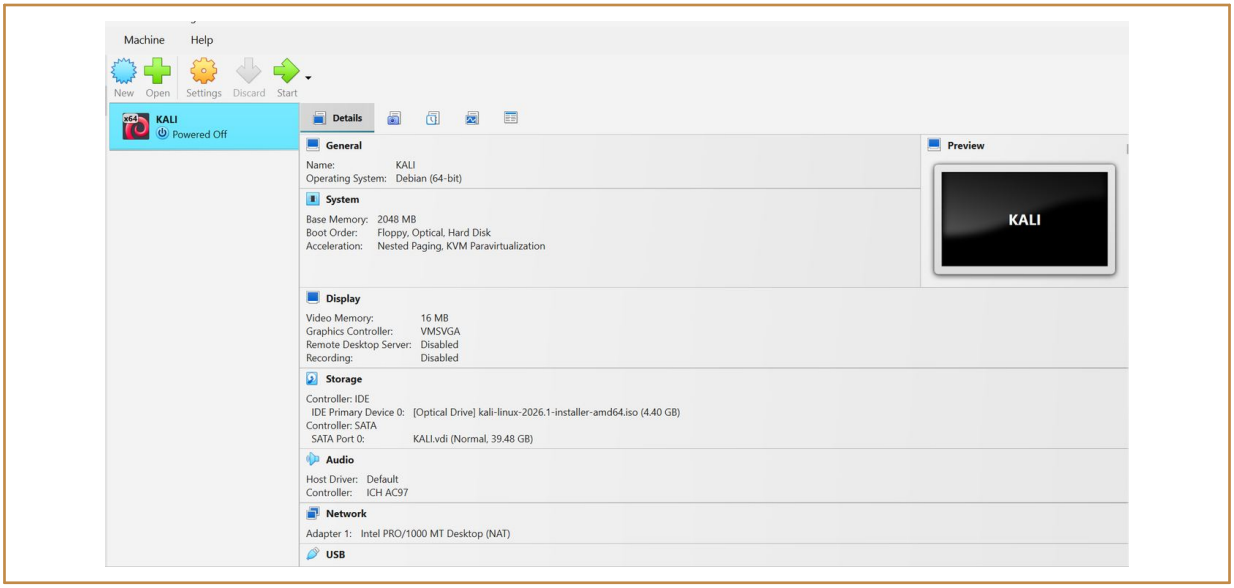


Figure — KALI machine configuration.

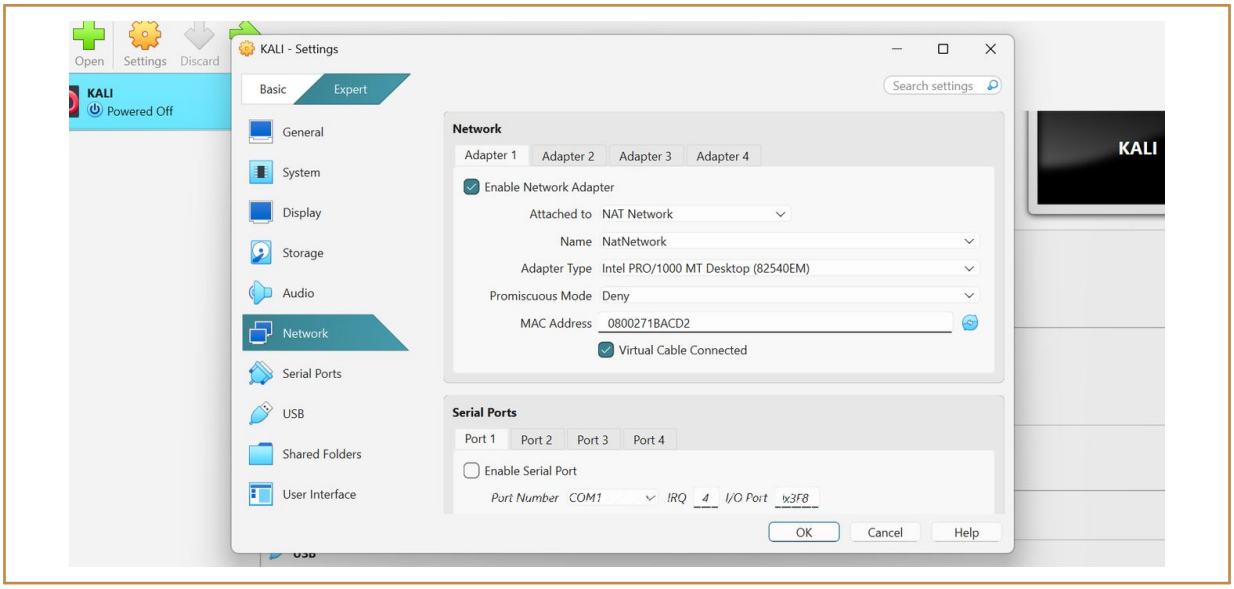


Figure — NAT Network adapter details.

04 | STARTING THE KALI INSTALLATION

Kali Linux was started from the virtual machine's boot media. The captured screens show the installer opening and continuing through the early network configuration portion of the installation.

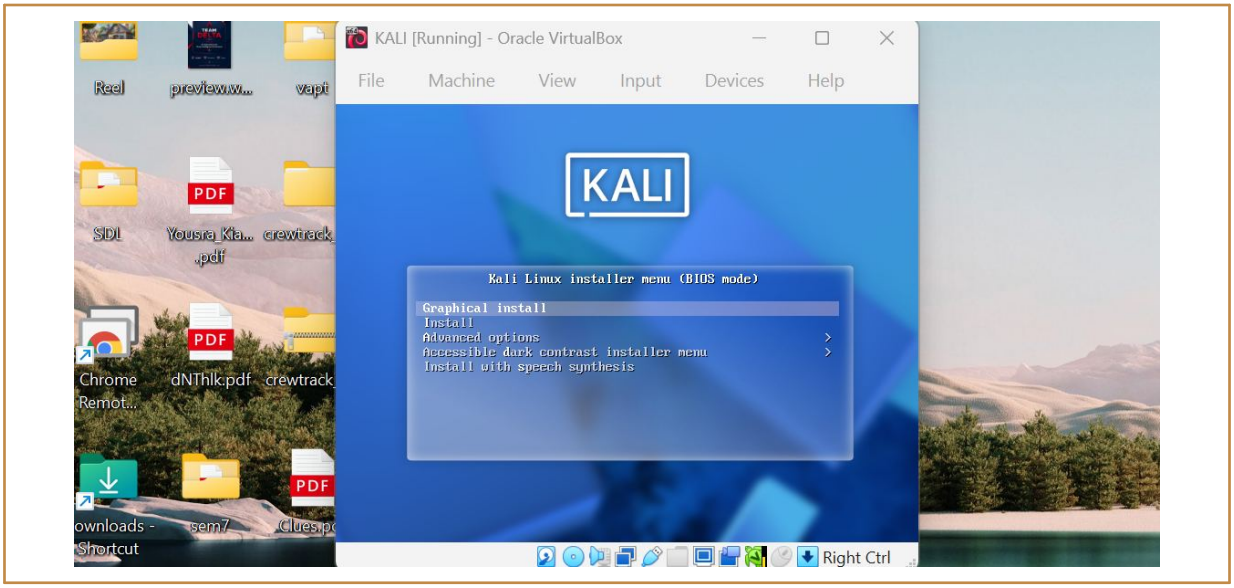


Figure — Kali Linux installer screen.

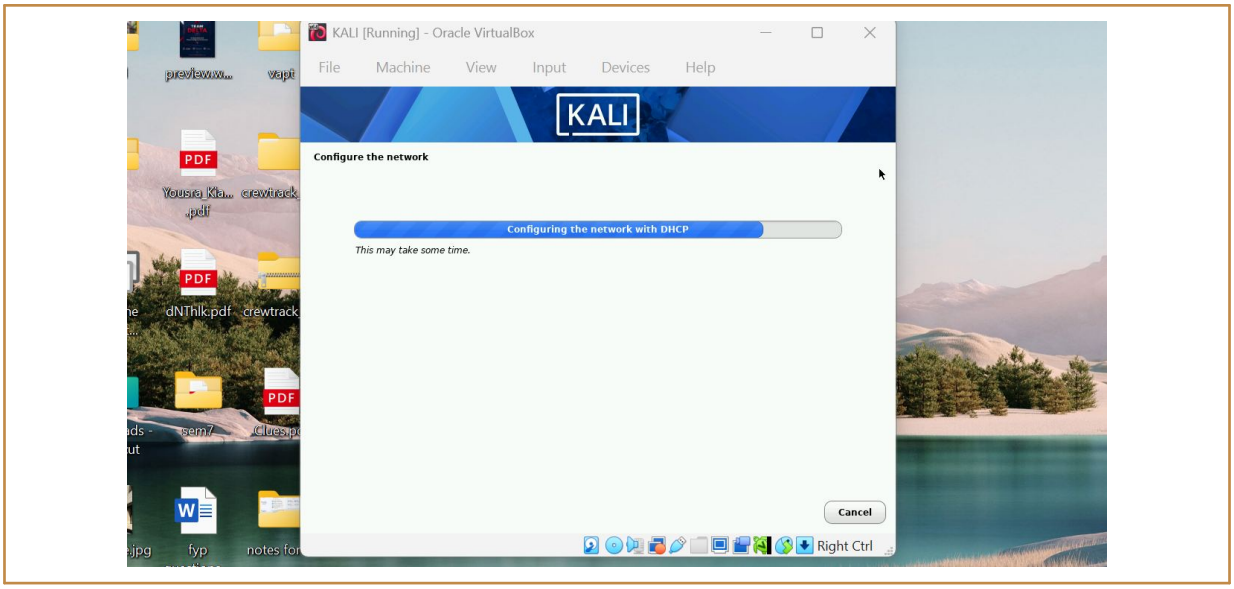


Figure — Network setup during installation.

05 | CONFIRMING NETWORK CONFIGURATION AND CONNECTIVITY

The Kali guest was assigned the required 10.0.0.2/24 address. A separate terminal ping test returned replies from the tested network address, with 10 responses received. The accompanying Kali desktop capture shows the wired network connection menu. These images provide evidence of the configured interface and the connectivity test carried out during the lab.

```
┌──(kali㉿kali)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.0.2  netmask 255.255.255.0  broadcast 10.0.0.255
        inet6 fe80::a00:27ff:fe1b:acd2  prefixlen 64  scopeid 0×20<link>
        ether 08:00:27:1b:ac:d2  txqueuelen 1000  (Ethernet)
        RX packets 1  bytes 590 (590.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 27  bytes 3070 (2.9 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0×10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Figure — Kali interface output showing 10.0.0.2/24.

```
┌──(kali㉿kali)-[~]
└─$ ping 192.168.0.123
PING 192.168.0.123 (192.168.0.123) 56(84) bytes of data.
64 bytes from 192.168.0.123: icmp_seq=1 ttl=64 time=1171 ms
64 bytes from 192.168.0.123: icmp_seq=2 ttl=64 time=146 ms
64 bytes from 192.168.0.123: icmp_seq=3 ttl=64 time=16.0 ms
64 bytes from 192.168.0.123: icmp_seq=4 ttl=64 time=185 ms
64 bytes from 192.168.0.123: icmp_seq=5 ttl=64 time=413 ms
64 bytes from 192.168.0.123: icmp_seq=6 ttl=64 time=288 ms
64 bytes from 192.168.0.123: icmp_seq=7 ttl=64 time=253 ms
64 bytes from 192.168.0.123: icmp_seq=8 ttl=64 time=174 ms
64 bytes from 192.168.0.123: icmp_seq=9 ttl=64 time=299 ms
64 bytes from 192.168.0.123: icmp_seq=10 ttl=64 time=424 ms
^C
── 192.168.0.123 ping statistics ──
11 packets transmitted, 10 received, 9.09091% packet loss, time 10037ms
rtt min/avg/max/mdev = 16.017/336.835/1171.478/301.659 ms, pipe 2
```

Figure — Ping test output: 10 replies received.



Figure — Kali desktop with wired network connection menu.

06 | MAKING THE HOST DOWNLOADS FOLDER AVAILABLE

The Downloads directory on the Windows host was added to the KALI virtual machine through the Shared Folders settings. Auto-mount was enabled, and the Kali file manager shows the shared location from inside the guest operating system.

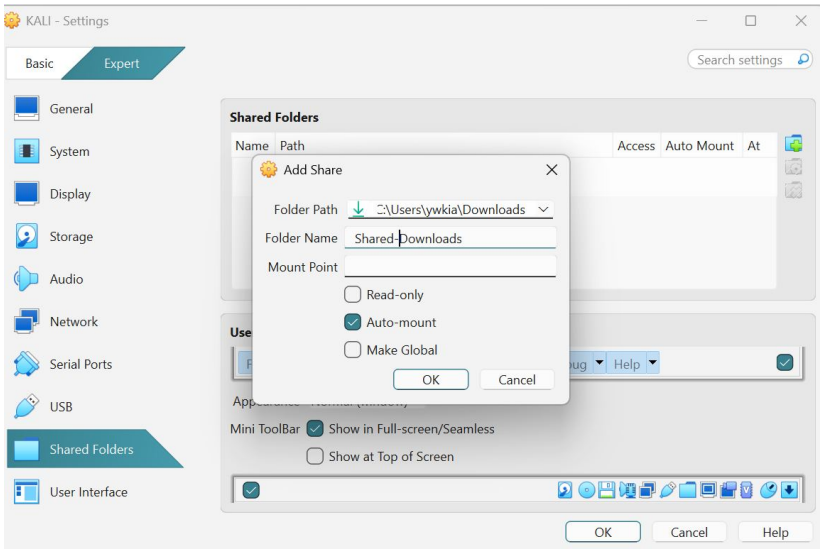


Figure — Shared folder setup under the yw-kiani host profile.

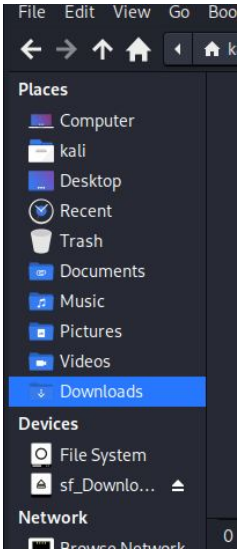


Figure — Downloads location in the Kali file manager.

07 | FINAL VERIFICATION AND BACKUP POINT

After the installation and configuration stages, the lab was checked against the task requirements. Clipboard and drag-and-drop integration were enabled, Internet connectivity was confirmed from Kali Linux, and a virtual-machine snapshot was saved to provide a restore point. The required setup was therefore completed.

FINAL RESULT

The VirtualBox and Kali Linux lab was completed against the task requirements. The guest was placed on the required NAT Network and configured with 10.0.0.2/24. The Downloads directory was shared with the guest, clipboard and drag-and-drop integration were enabled, Internet access was confirmed, and a VM snapshot was

saved as a restore point.

The screenshots in this report document the setup stages and the network and shared-folder checks performed during the practical.